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标题	Transition metal-free visible light photoredox-catalyzed remote C(sp3) − H borylation enabled by 1,5-hydrogen atom transfer
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摘要（原文节选）

The borylation of unreactive carbon-hydrogen bonds is a valuable method for transforming feedstock chemicals into versatile building blocks. Here, we describe a transition metal-free method for the photoredox-catalyzed borylation of unactivated C(sp3) − H bond, initiated by 1,5-hydrogen atom transfer (HAT). The remote borylation was directed by 1,5-HAT of the amidyl radical, which was generated by photocatalytic reduction of hydroxamic acid derivatives. The method accommodates substrates with primary, secondary and t ⋯

数据表 · Table 1（20 行，含数值行 20 行）

Optimization of reaction conditionsa.

Entry	B2cat2	PC (mol%)	Light	Additive	Yielda,b(%)
1	3.0	Ir(p-CF3ppy)3 (2 mol%)	456 nm	Et3N (2.0)	45
2	3.0	Ir(p-CF3ppy)3 (2 mol%)	456 nm	DABCO (2.0)	15
3	3.0	Ir(p-CF3ppy)3 (2 mol%)	456 nm	DIPEA (2.0)	47
4	3.0	Ir(p-CF3ppy)3 (2 mol%)	456 nm	none	trace
5c	3.0	Ir(p-CF3ppy)3 (2 mol%)	456 nm	DIPEA (2.0)	49
6	3.0	Ir(ppy)3 (2 mol%)	456 nm	DIPEA (2.0)	30
7	3.0	[Ru(bpy)3]Cl2 (2 mol%)	456 nm	DIPEA (2.0)	trace
8	3.0	4CzIPN (2 mol%)	456 nm	DIPEA (2.0)	trace
9	3.0	Eosin Y (2 mol%)	456 nm	DIPEA (2.0)	47
10	3.0	Eosin Y (2 mol%)	467 nm	DIPEA (2.0)	73
11	3.0	Eosin Y (5 mol%)	467 nm	DIPEA (2.0)	77
12	2.5	Eosin Y (5 mol%)	467 nm	DIPEA (2.0)	81 (73)
13	4.0	Eosin Y (5 mol%)	467 nm	DIPEA (2.0)	40
14	2.5	Eosin Y (5 mol%)	467 nm	DIPEA (1.5)	54
15	2.5	None	467 nm	DIPEA (2.0)	trace
16	2.5	Eosin Y (5 mol%)	none	DIPEA (2.0)	trace
17	2.5	Eosin Y (5 mol%)	467 nm	None	trace
18d	2.5	Eosin Y (5 mol%)	none	DIPEA (2.0)	trace
19e	2.5	Eosin Y (5 mol%)	467 nm	DIPEA (2.0)	trace
20f	2.5	Eosin Y (5 mol%)	467 nm	DIPEA (2.0)	76

脚注（逐字）：aReactions run on 0.2 mmol scale, using Kessil 40 W blue LED as light source.bGC-fid yields are measured by using decane as an internal standard. The yield in bracket is isolated yield.cThe reaction time is 30 h.d50 ° C for 12 h, in the dark.eTurned off the light after 30 min.fThe reaction was set up

正文对应句（逐字）：Recently, visible-light photoredox catalysis has been the leading efficient access to N-radicals under mild conditions. With their weak N − O bond, hydroxylamine derivatives have recently been exploited as nitrogen-centered radical precursors in visible-light photocatalysis, given their higher oxidati

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